College of Engineering and Technology

Project Work – Student Hand Book

Project Batch ID

Degree/ program	B.Tech	**Specialisation**	Computer Science & Engineering
Academic Year	2025-2026 (Even)	**Semester**	6

Name of student	**Register Number**	**Department**	**Mobile Number**	**Email ID**
Girijesh B	RA2311030010286	NWC	6379883776	bt1368@srmist.edu.in
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Title of the Project:	**A UNIFIED AI-BASED NETWORK INTRUSION DETECTION FRAMEWORK FOR NETWORK AND SYSTEM TRAFFIC**
Project Site / Location	
Name and address of the company / organisation (Applicable for projects with industry or industry support)	SRM IST, Kattankulathur, Chengalpattu District-603203

Supervision Team

	Supervisor	**Co-Supervisor**	**External Supervisor (If applicable)**
Name	Dr.B. Yamini		
Designation	Associate Professor		
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Course Code	21CSP302L	Course Title	Project

Mission Statement

To design and develop a lightweight, intelligent, and real-time AI-based Intrusion Detection System integrated with a Security Operations Center (SOC) dashboard for securing enterprise using machine learning, anomaly detection, and graph-based threat analysis.

Problem (or) Product Description:

Modern networks face attacks such as malware, zero-day exploits, insider threats, and unauthorized access. Traditional IDS cannot detect advanced attacks effectively and often generate too many alerts.

This project proposes an AI-Based IDS for SOC using Isolation Forest, Random Forest, GNN, and Hybrid Command Detection. It captures live traffic, detects threats in real time, and displays alerts through a web dashboard.

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Assumptions and Constraints

Assumptions: Real-time network data is available and Training data contains normal and attack traffic.

Constraints: Must work with low latency, Limited system resources and False positives should be minimized.

Stakeholders

SOC Analysts, Network Administrators, Organizations, Cybersecurity Teams.

Division of work and contributors of SPRINT 1 [Include Daily Scrum of Sprint 1]

Sprint Duration: 5/1/2026 – 31/1/2026

Team Members: Gowtham, Girijesh, Shyam

Sprint 1 focused on problem analysis, literature survey, dataset collection, and initial system design for the AI-Based Intrusion Detection System with SOC Dashboard. This sprint established the project foundation.

Work Distribution:

Gowtham – Research & System Design

- Conducted literature survey on IDS, SOC, and AI-based security systems
- Studied attacks such as DoS, brute force, malware, and insider threats
- Analyzed existing IDS limitations and research gaps
- Defined objectives and system architecture
- Helped select hybrid AI detection approach

Girijesh – Data Collection & Preprocessing

- Collected network traffic datasets for training/testing
- Cleaned noisy and incomplete records
- Converted raw traffic into structured flow format
- Applied feature extraction and normalization
- Prepared labeled datasets for model training

Shyam – Environment Setup & Initial Implementation

- Set up Python, Flask, database, and required libraries
- Organized project folders and code modules
- Developed initial packet capture scripts using Scapy
- Started basic Isolation Forest implementation
- Planned integration for dashboard and future modules

Daily Scrum of Sprint 1

Day 1 – Sprint Planning
- Finalized project title: AI-Based IDS for SOC Environments
- Defined scope, objectives, and expected outcomes
- Assigned roles to team members

Day 2 – Problem Understanding
- Studied enterprise and cloud network security issues
- Identified IDS limitations and alert overload problems
- Discussed real-world cyber threats

Day 3 – Literature Survey
- Reviewed research papers on IDS techniques
- Compared statistical, ML, and DL approaches
- Identified limitations of existing systems

Day 4 – Dataset Collection
- Collected network traffic datasets
- Explored features, labels, and traffic structure
- Identified preprocessing needs

Day 5 – Data Cleaning
- Removed missing and inconsistent values
- Converted raw logs into usable format
- Performed initial feature selection

Day 6 – Data Preprocessing
- Applied normalization and scaling
- Converted labels into binary classification
- Structured dataset for model input

Day 7 – Pipeline Validation
- Verified preprocessing pipeline output
- Ensured compatibility with ML models
- Fixed minor data inconsistencies

Day 8 – Environment Setup

- Installed required libraries (NumPy, Pandas,TensorFlow)
- Configured development environment
- Organized project folders

Day 9 – Initial Model Setup

- Started basic implementation of Isolation Forest
- Connected preprocessed data with model input
- Tested initial execution

Day 10 – Review & Documentation

- Reviewed Sprint 1 progress
- Documented dataset and preprocessing steps
- Identified challenges (data imbalance, feature selection)
- Planned next sprint for model development (IF, AE, LSTM)

Signature of the Supervisor

Division of work and contributors of SPRINT 2 [Include Daily Scrum of SPRINT 2]

Sprint Duration: 1/2/2026 – 28/2/2026
Team Members: Gowtham, Girijesh, Shyam

Sprint 2 focused on model development, training, and evaluation of individual AI algorithms (Isolation Forest, Random Forest, and Graph Neural Network). The goal of this sprint was to implement and test each model independently for intrusion detection.

Work Distribution

Gowtham – Isolation Forest (Statistical Model)

- Implemented **Isolation Forest** for anomaly detection
- Trained model using normal traffic data
- Tuned parameters such as contamination and estimators
- Detected anomalies based on traffic patterns (DoS, abnormal frequency)
- Evaluated model performance and recorded results

Girijesh – Autoencoder (Payload-Based Detection)

- Designed and implemented **Autoencoder neural network**
- Trained model using normal dataset to learn data patterns
- Calculated **reconstruction error** for anomaly detection
- Set threshold (mean + standard deviation)
- Detected payload-level anomalies and unknown attacks

Shyam – LSTM (Temporal Detection)

- Developed **LSTM model** for sequence-based anomaly detection
- Generated time-series sequences from dataset
- Trained model to learn temporal dependencies
- Detected replay attacks and timing inconsistencies
- Evaluated sequence prediction accuracy

Daily Scrum of Sprint 2

Day 1 – Sprint Planning

- Defined Sprint 2 goal: model development (IF, AE, LSTM)
- Assigned model responsibilities to each member
- Reviewed preprocessed dataset from Sprint 1

Day 2 – Isolation Forest Setup

- Started implementation of Isolation Forest
- Prepared training data (normal traffic only)
- Configured initial parameters

Day 3 – Autoencoder Development

- Designed Autoencoder architecture (encoder + decoder)
- Began training using normal dataset
- Monitored reconstruction loss

Day 4 – LSTM Implementation

- Developed LSTM model structure
- Converted data into time sequences
- Began training for temporal learning

Day 5 – Model Training & Tuning

- Continued training of all models
- Tuned Autoencoder parameters (epochs, threshold)
- Improved Isolation Forest accuracy

Day 6 – Model Evaluation

- Tested Isolation Forest results (traffic anomalies)
- Evaluated Autoencoder reconstruction error
- Tested LSTM for sequence anomalies

Day 7 – Comparison & Analysis

- Compared outputs of all three models
- Identified strengths:
  - IF → traffic anomalies
  - AE → payload anomalies
  - LSTM → sequence anomalies
- Documented results

Day 8 – Optimization

- Improved model performance through tuning
- Reduced false positives
- Balanced accuracy and efficiency

Day 9 – Validation

- Verified model outputs with test dataset
- Checked consistency across different scenarios
- Ensured models are ready for integration

Day 10 – Sprint Review & Planning

- Reviewed overall Sprint 2 progress
- Identified challenges (training time, tuning)
- Planned Sprint 3 → Hybrid Integration + Final Evaluation

Signature of the Supervisor

Division of work and contributors of SPRINT 3 [Include Daily Scrum of Sprint 3]

Sprint Duration: 1/3/2026 – 31/3/2026
Team Members: Gowtham, Girijesh, Shyam

Sprint 3 focused on integration of all models into a Hybrid Intrusion Detection System, performance evaluation, and final system validation. This sprint ensured that the system works in real-time and produces accurate results.

Work Distribution

Gowtham – Hybrid Integration & Fusion Logic

- Designed hybrid architecture combining IF, Autoencoder, and LSTM
- Implemented fusion logic (if any model detects anomaly → attack)
- Integrated outputs of all models into a unified system
- Ensured smooth data flow between modules
- Worked on system-level testing

Girijesh – Performance Evaluation & Metrics

- Evaluated system using:
    - Accuracy
    - Precision
    - Recall
    - F1-score
- Compared hybrid model with individual models
- Analyzed false positives and false negatives

- Generated performance reports and tables

Shyam – Real-Time Simulation & Final Implementation

- Simulated real-time data flow
- Tested system under different attack scenarios
- Optimized model execution for faster response
- Ensured system runs efficiently on companion computer
- Prepared final outputs and result visualization

Daily Scrum of Sprint 3:

Day 1 – Sprint Planning

- Defined Sprint 3 goal: hybrid integration and final evaluation
- Discussed integration strategy for three models
- Assigned responsibilities

Day 2 – Integration Setup

- Connected outputs of Isolation Forest, Autoencoder, and LSTM
- Designed data flow between modules
- Began implementation of hybrid structure

Day 3 – Fusion Logic Implementation

- Implemented decision logic:
    - If any model detects anomaly → classify as attack
- Tested basic hybrid output

- Verified integration correctness

Day 4 – System Testing

- Tested hybrid system on dataset
- Verified detection of multiple attack types
- Checked consistency of results

Day 5 – Performance Evaluation

- Calculated accuracy, precision, recall, F1-score
- Compared hybrid model with individual models
- Observed performance improvement

Day 6 – Real-Time Simulation

- Simulated real-time message flow
- Tested system latency and response time
- Verified real-time anomaly detection capability

Day 7 – Optimization

- Reduced computational overhead
- Improved model execution speed
- Balanced accuracy vs efficiency

Day 8 – Result Analysis

- Analyzed detection results for different attacks
- Evaluated strengths of hybrid approach
- Documented observations

Day 9 – Final Validation

- Tested system with multiple scenarios
- Ensured reliability and robustness
- Verified final outputs

Day 10 – Documentation & Review

- Prepared final report content
- Created diagrams, tables, and results
- Reviewed complete project
- Prepared for presentation and viva

Signature of the Supervisor

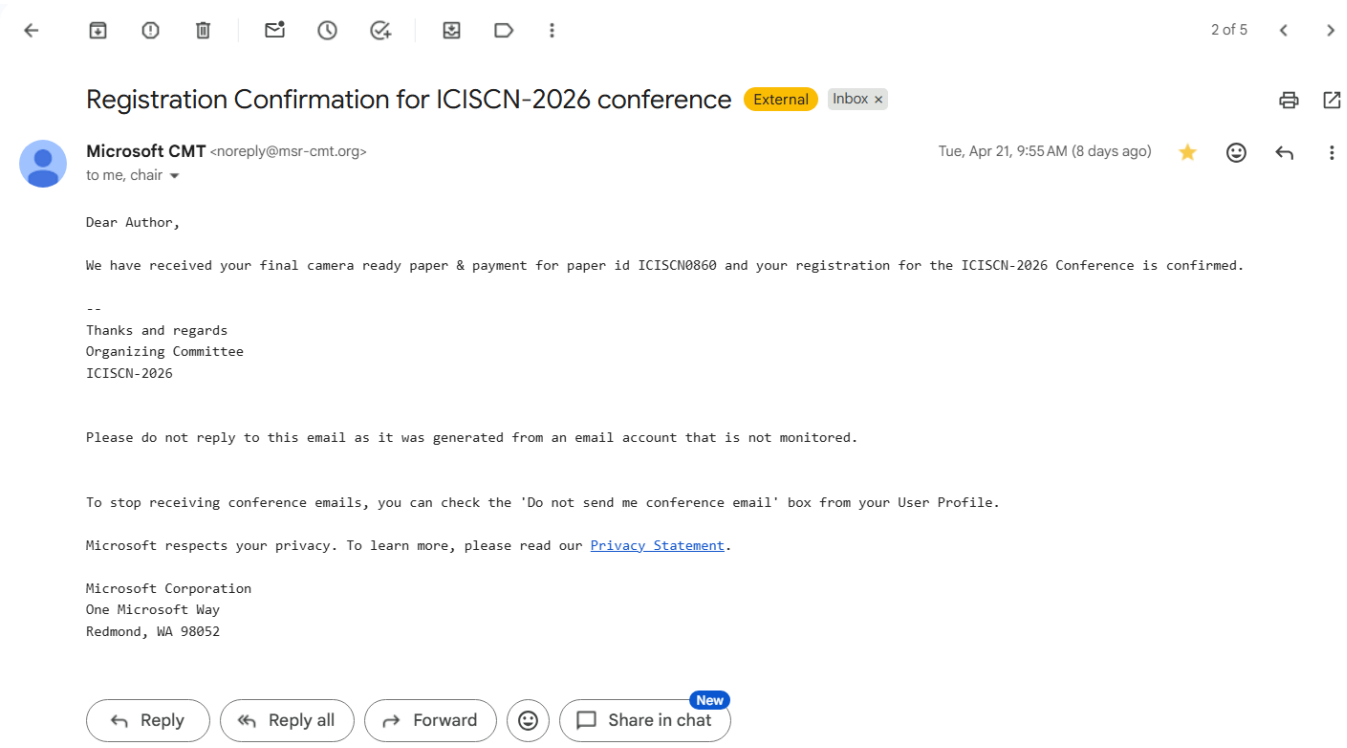
Worksheet / Data collection / Observation etc

Phase	Activity	Description
Data Understanding	Dataset Study	dataset was analyzed to understand structure, message format, ID distribution, and payload characteristics
	Feature Identification	Important features such as timestamp, ID, and payload bytes were identified
	Target Definition	Classification target defined as Normal vs Attack traffic
	Data Exploration	Traffic patterns and anomaly distributions were analyzed
Data Preprocessing	Missing Values Handling	Missing or inconsistent frames were handled appropriately
	Noise Removal	Irrelevant and corrupted messages were removed
	Normalization	Features were scaled to a uniform range for model stability
	Data Cleaning	Duplicate and invalid entries were eliminated
Feature Engineering	Feature Extraction	Relevant features were extracted from messages
	Feature Selection	Important attributes selected for intrusion detection modeling
	Dimensionality Reduction	Redundant features removed to improve efficiency
	Final Feature Set	Optimized dataset prepared for training
Data Balancing	Imbalance Detection	Class imbalance identified between normal and attack data
	SMOTE Application	Synthetic samples generated for attack class
	Dataset Balancing	Balanced dataset achieved

Phase	Activity	Description
	Verification	Balanced dataset validated for correctness
Model Development	Model Selection	Isolation Forest, Autoencoder, and LSTM selected for detection
	Data Splitting	Dataset split into training and testing sets
	Training	Models trained using processed dataset
	Hyperparameter Tuning	Parameters optimized for better performance
Evaluation	Accuracy	Improved after model tuning
	Precision	Good classification performance observed
	Recall	Improved detection of attack instances
	F1-Score	Balanced performance achieved
	Final Observation	Hybrid IDS achieved improved detection accuracy

Research Article with Journal Publication Details / Patent disclosure form with patent status

(include certificates and proofs)



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